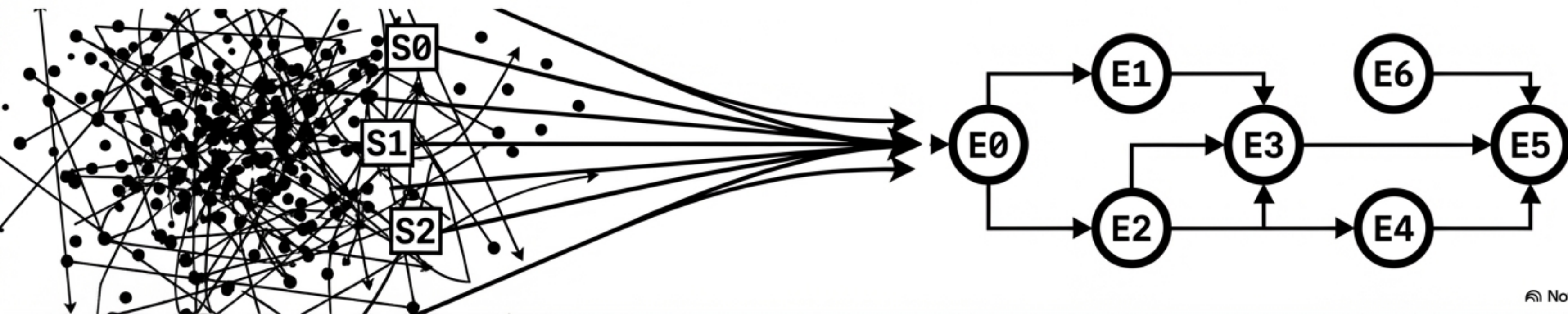


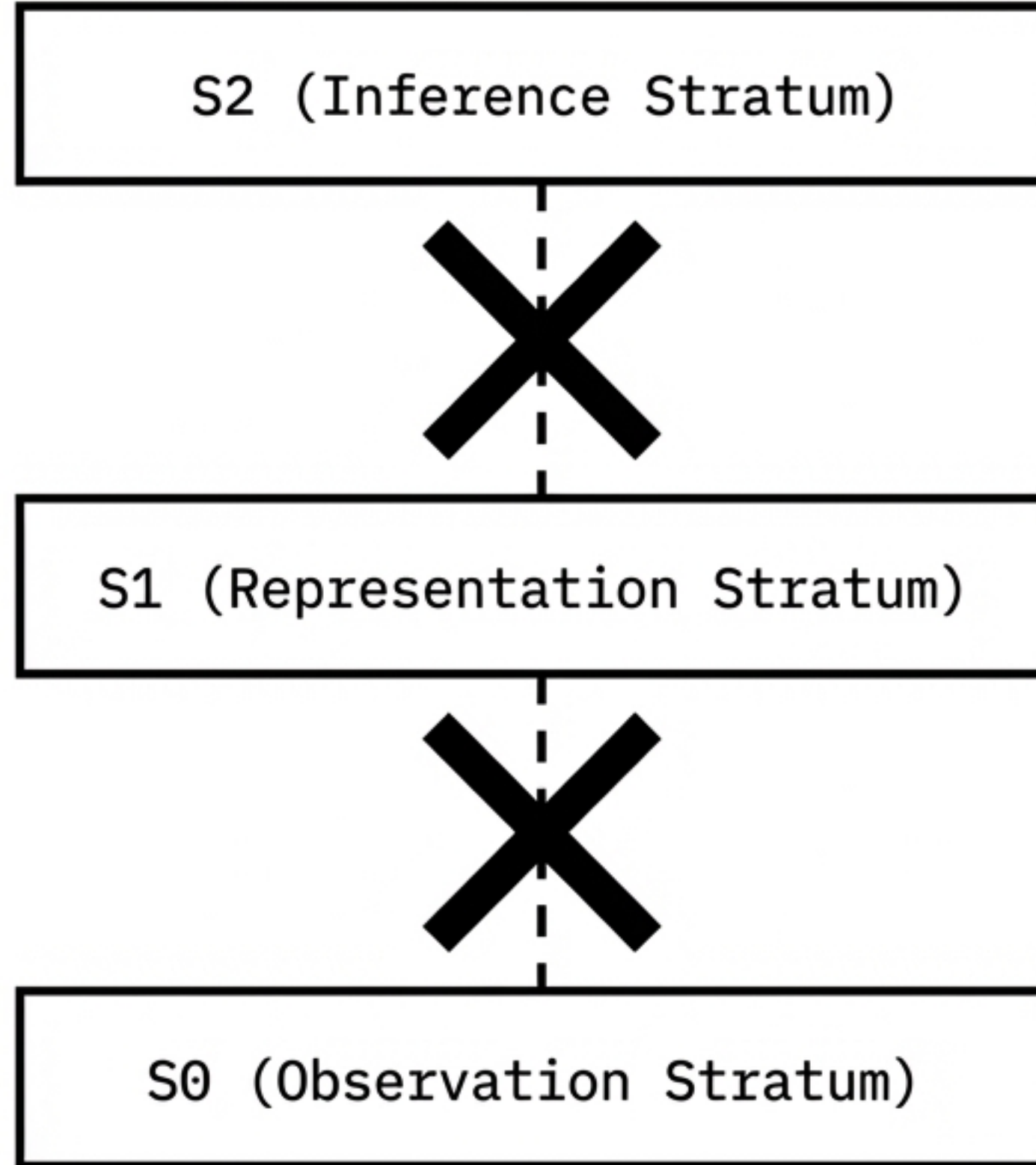
AMACE

Autonomous Multi-Stratum Anomaly Correlation Engine

The Strict Pipeline Architecture



The Cross-Stratum Blind Spot & Combinatorial Explosion



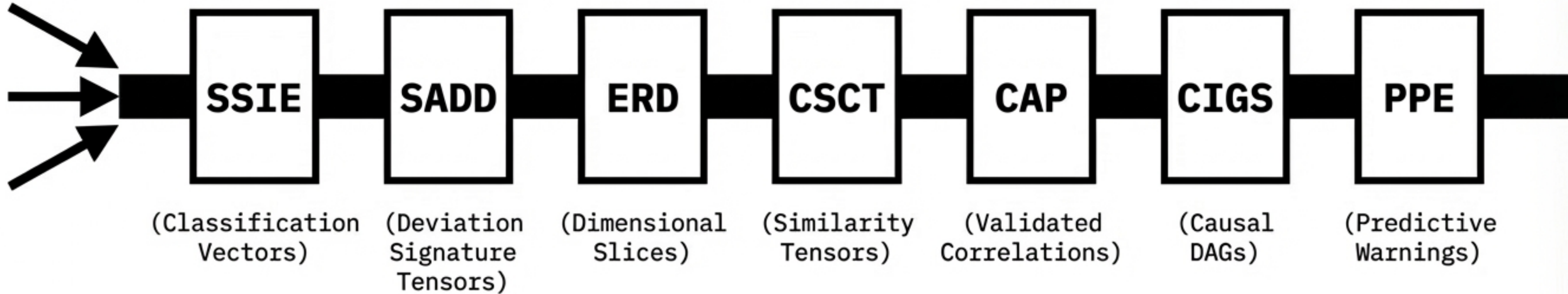
$$O(N^2 \times K^D) > 10^{18}$$

combinations

The mathematical impossibility of cross-stratum correlation using traditional exhaustive search methods.

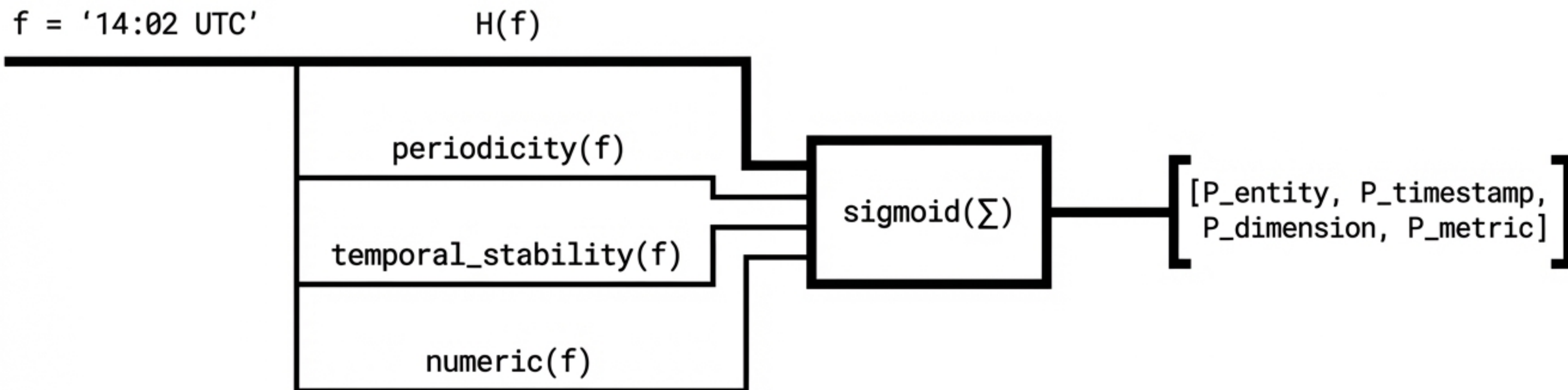
The Strict Pipeline: Macro Architecture

Raw Strata
(S0, S1, S2)

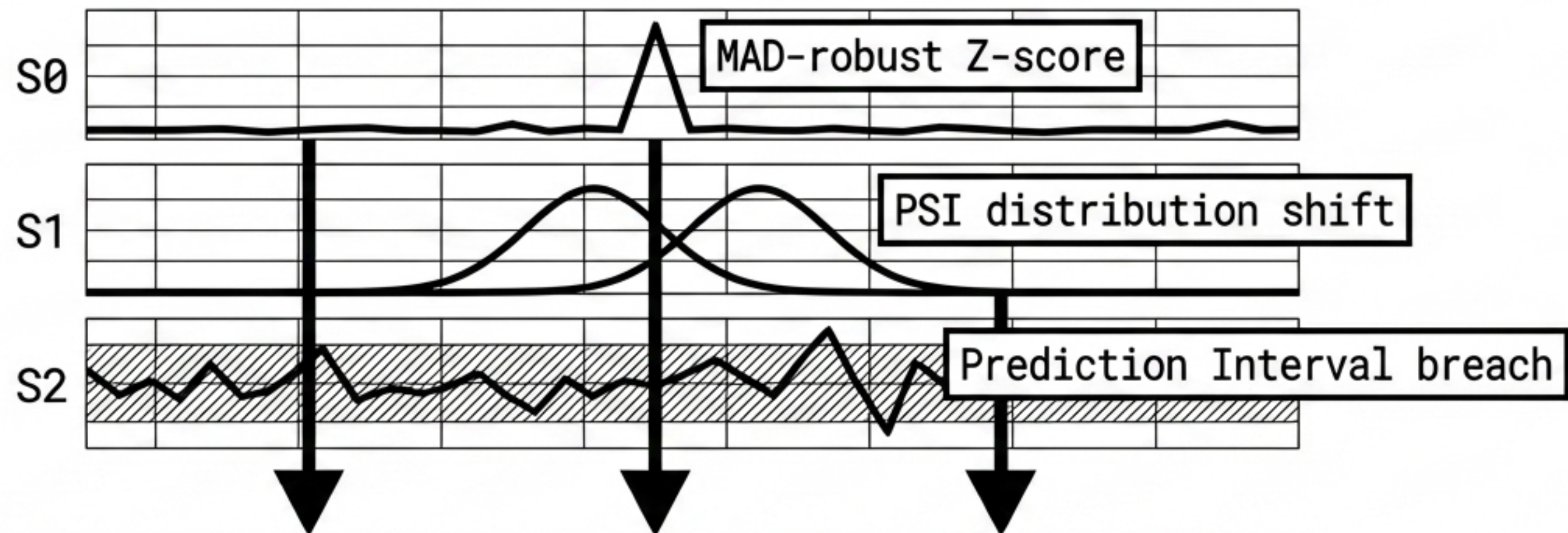


Input State [Raw Strata Data] -> SSIE -> Output Artifact [Classification Vectors]

Stratum Schema Inference Engine (SSIE)



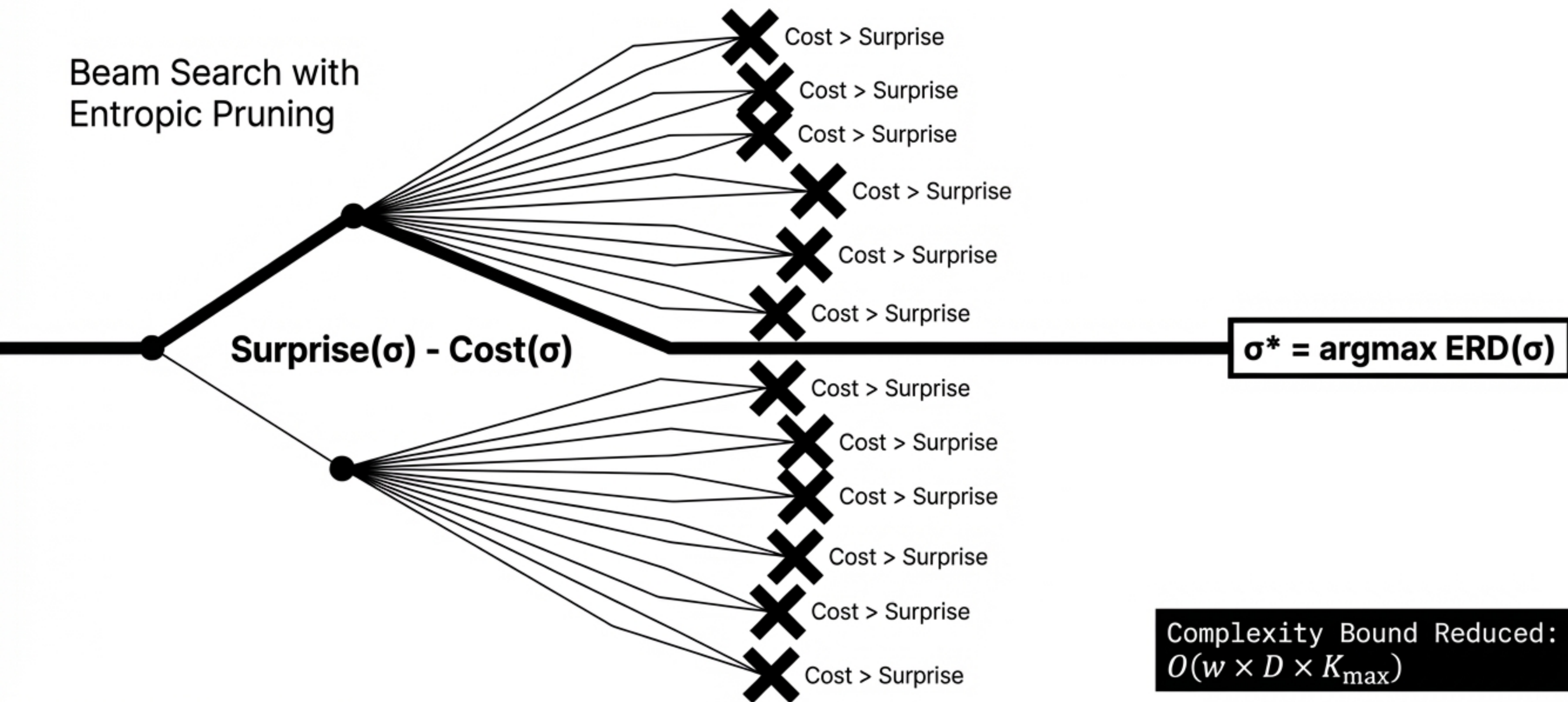
Stratum-Adaptive Deviation Detector (SADD)



$$DST = (S, E, T, D, M, C, \Phi)$$

S: Stratum Origin	M: Magnitude
E: Entity Set	C: Confidence
T: Temporal Envelope	Φ : Provenance Hash
D: Dimensional Locus	

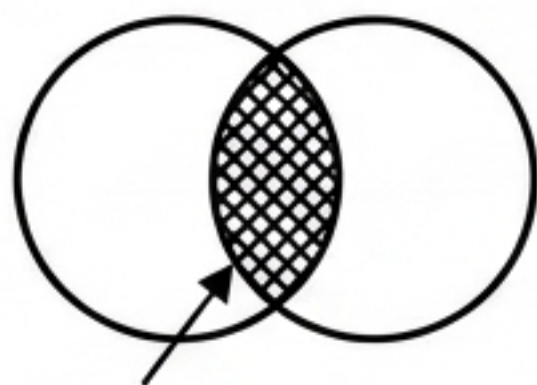
Entropic Relevance Decomposition: Dimensional Pruning



Cross-Stratum Causal Tensor: The Four Axes

Q1: Entity Coincidence

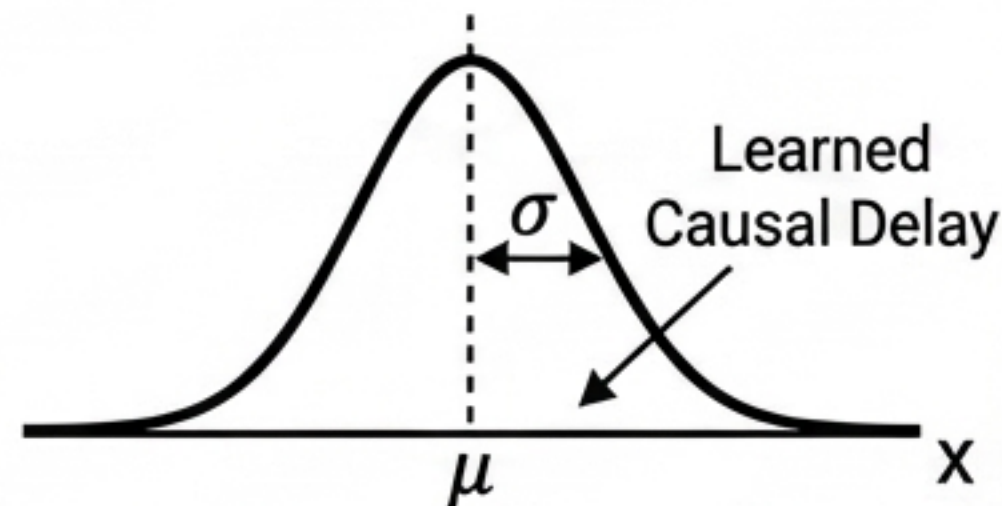
$$J(EA, EB) \times \lambda$$



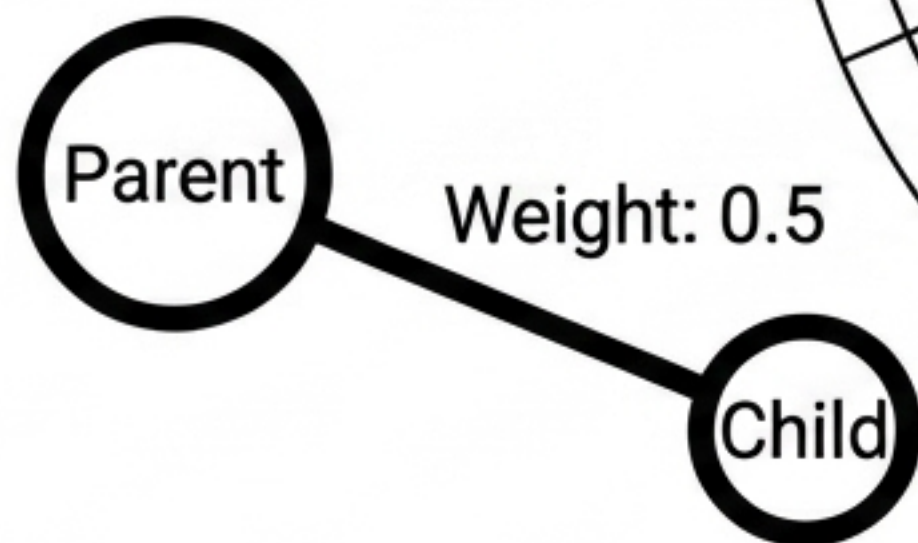
Surprise-adjusted Jaccard

Q2: Temporal Precedence

$$G(\text{onset_lag}, \mu, \sigma)$$

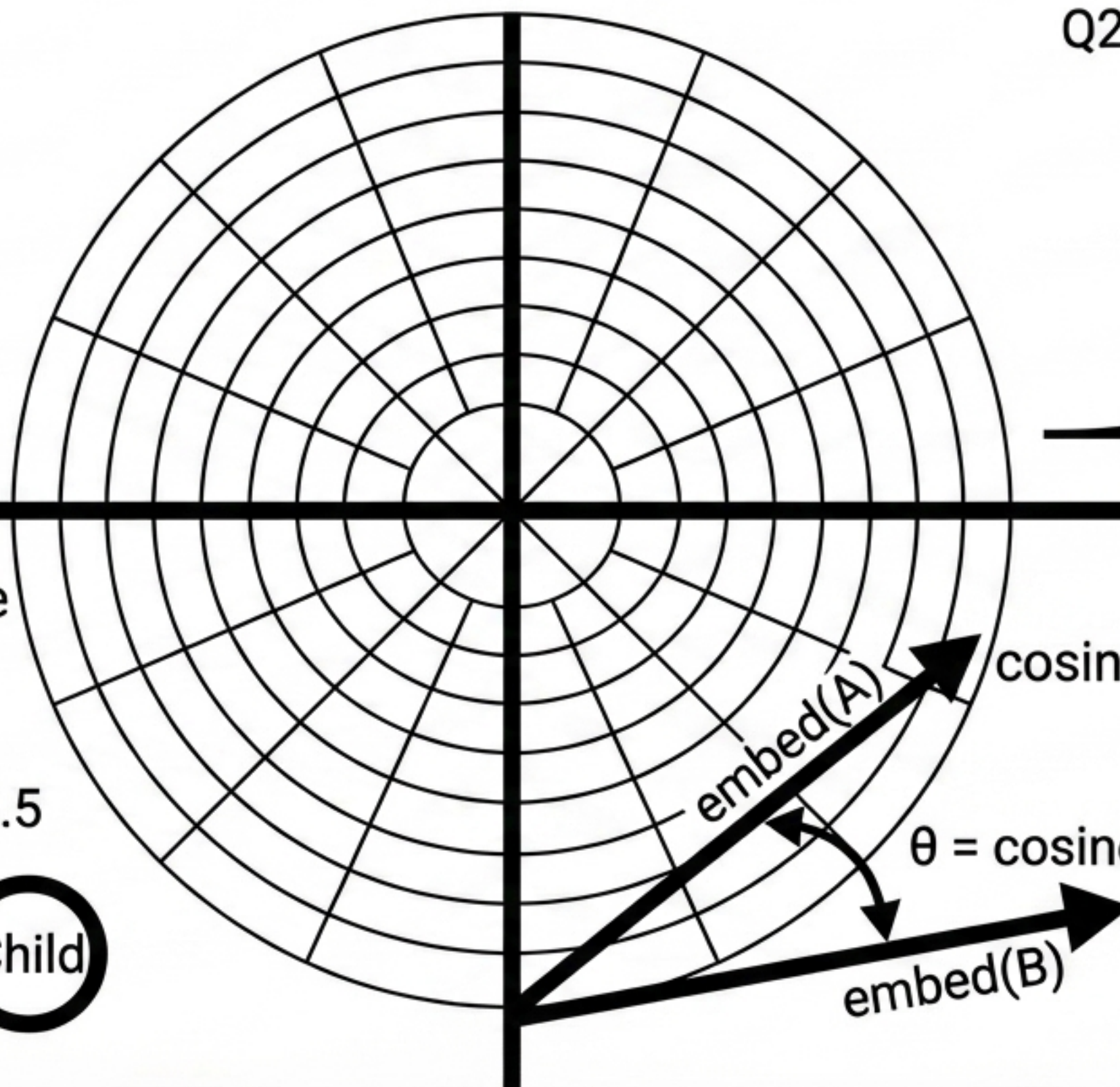


Q3: Dimensional Coherence



Q4: Semantic Affinity

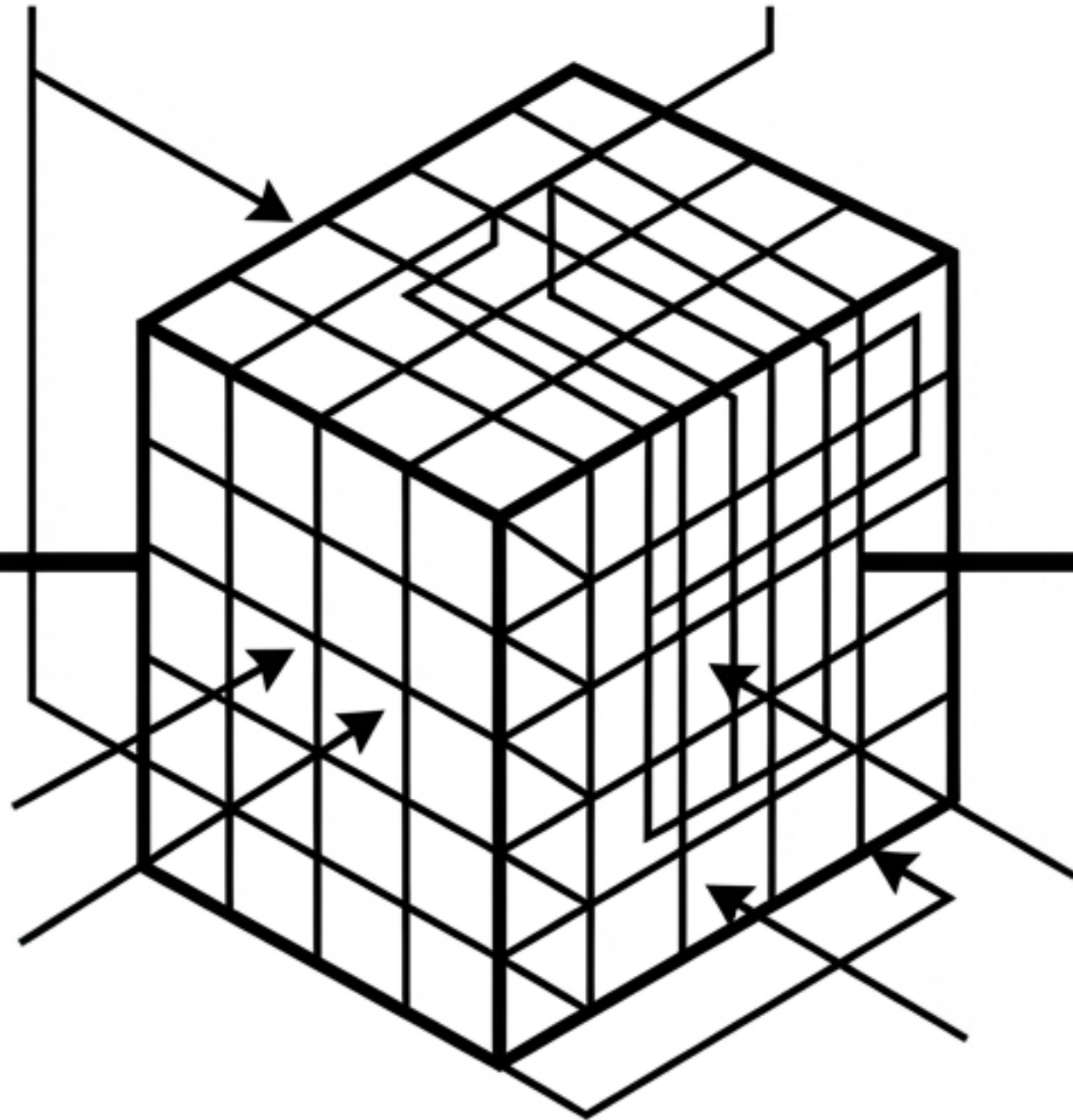
$$\text{cosine}(\text{embed}(A), \text{embed}(B))$$



Input State [4-Axis Similarity] -> CSCT -> Output Artifact [Similarity Tensors]

Spectral Normalization and Dynamic Weighting

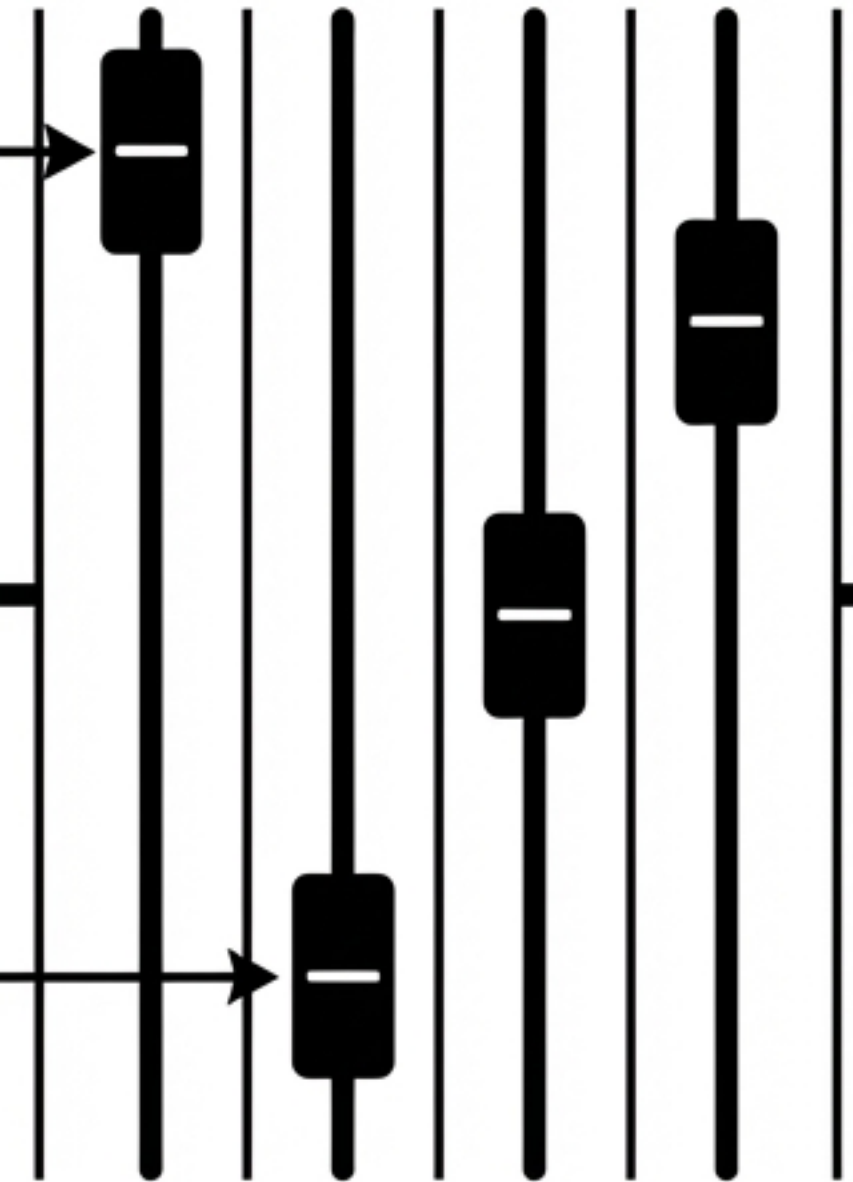
$$\mathbf{CSCT}(\mathbf{A}, \mathbf{B}) = \left\| \sum w_i \times \sigma_i(\mathbf{A}, \mathbf{B}) \times \hat{\mathbf{e}}_i \right\|$$



Dynamic Weighting Scale

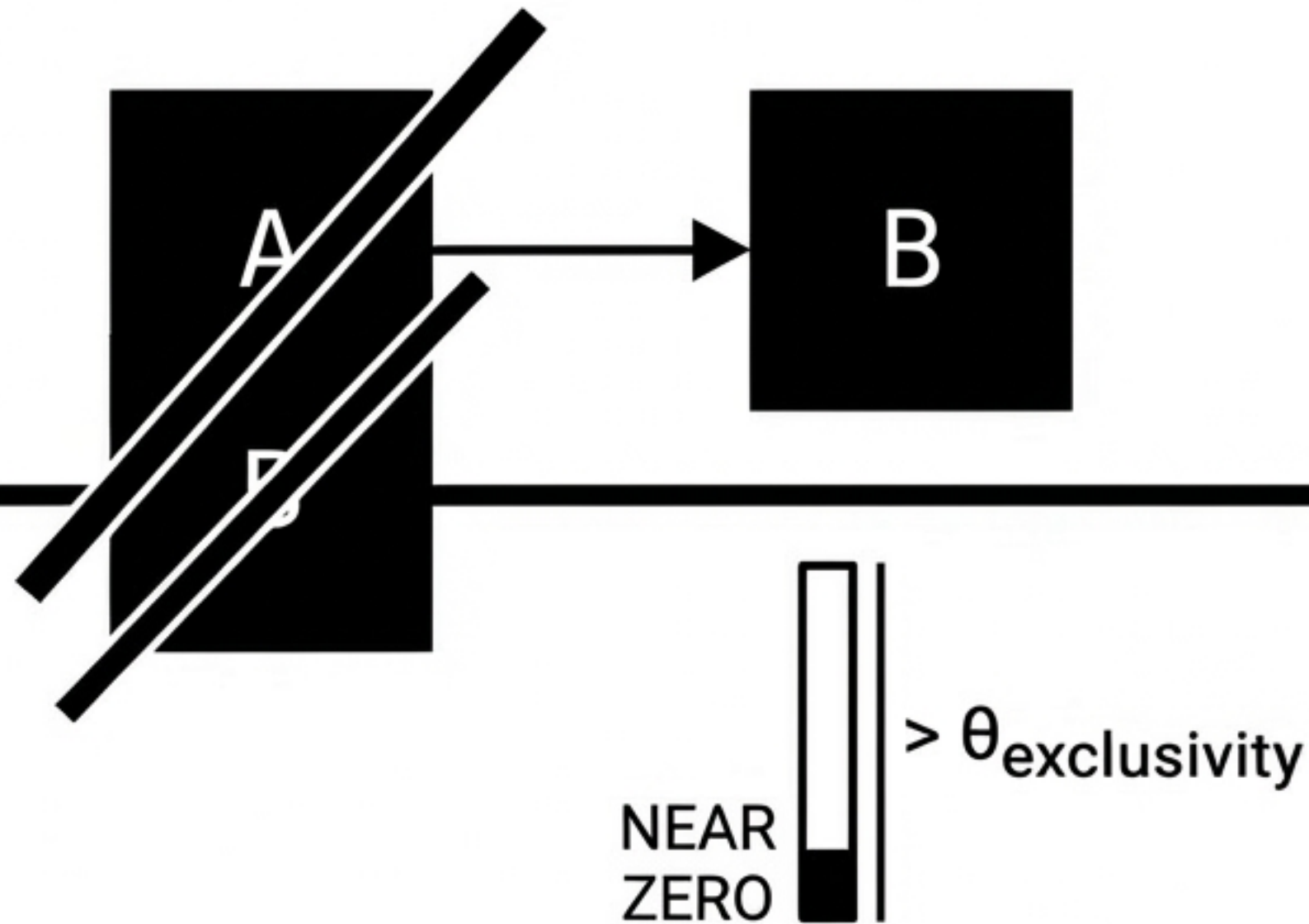
Sudden Onset
Detected ->
Temporal Weight
Shifted High

Gradual Drift
Detected ->
Entity Weight
Shifted High



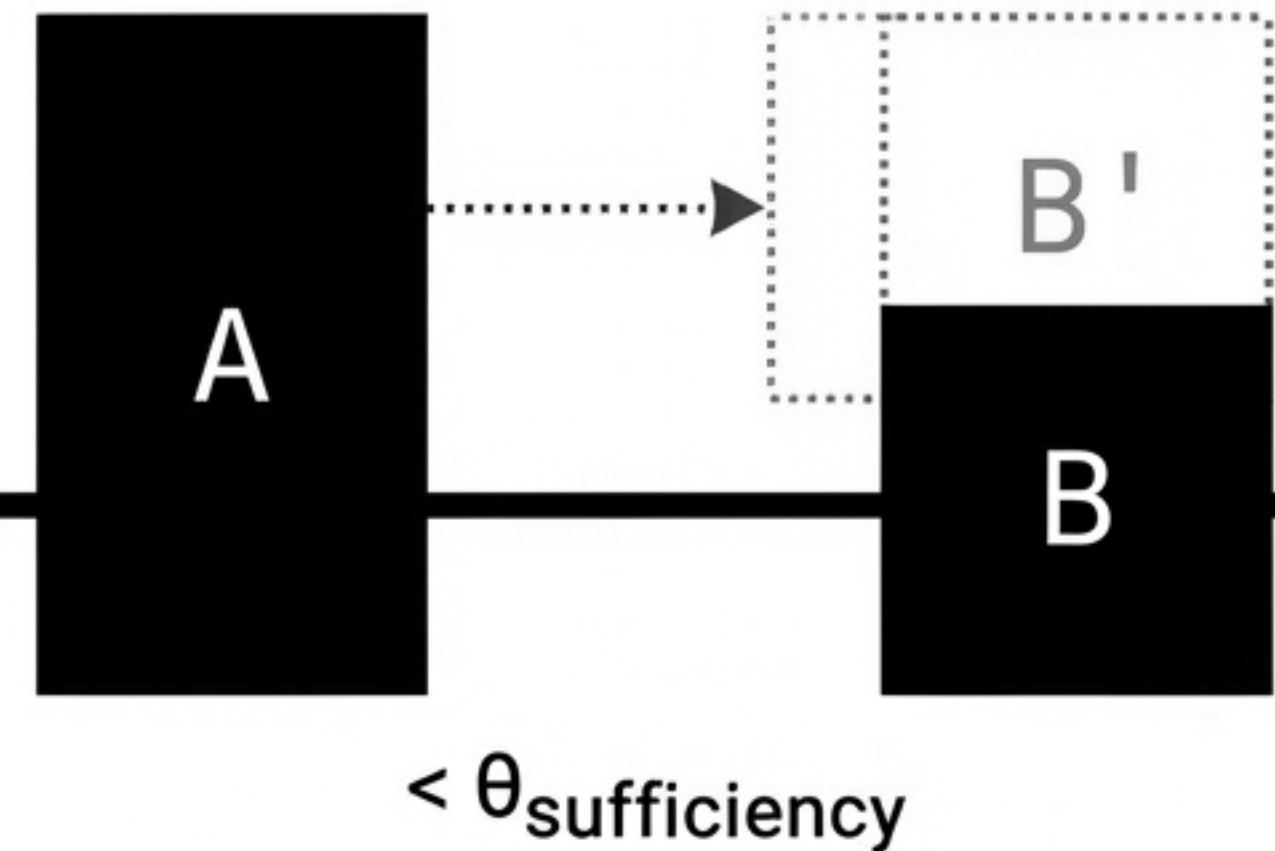
Computational Ablation Protocol: Exclusivity & Sufficiency

Test 1: Exclusivity



PASSED: A is Necessary

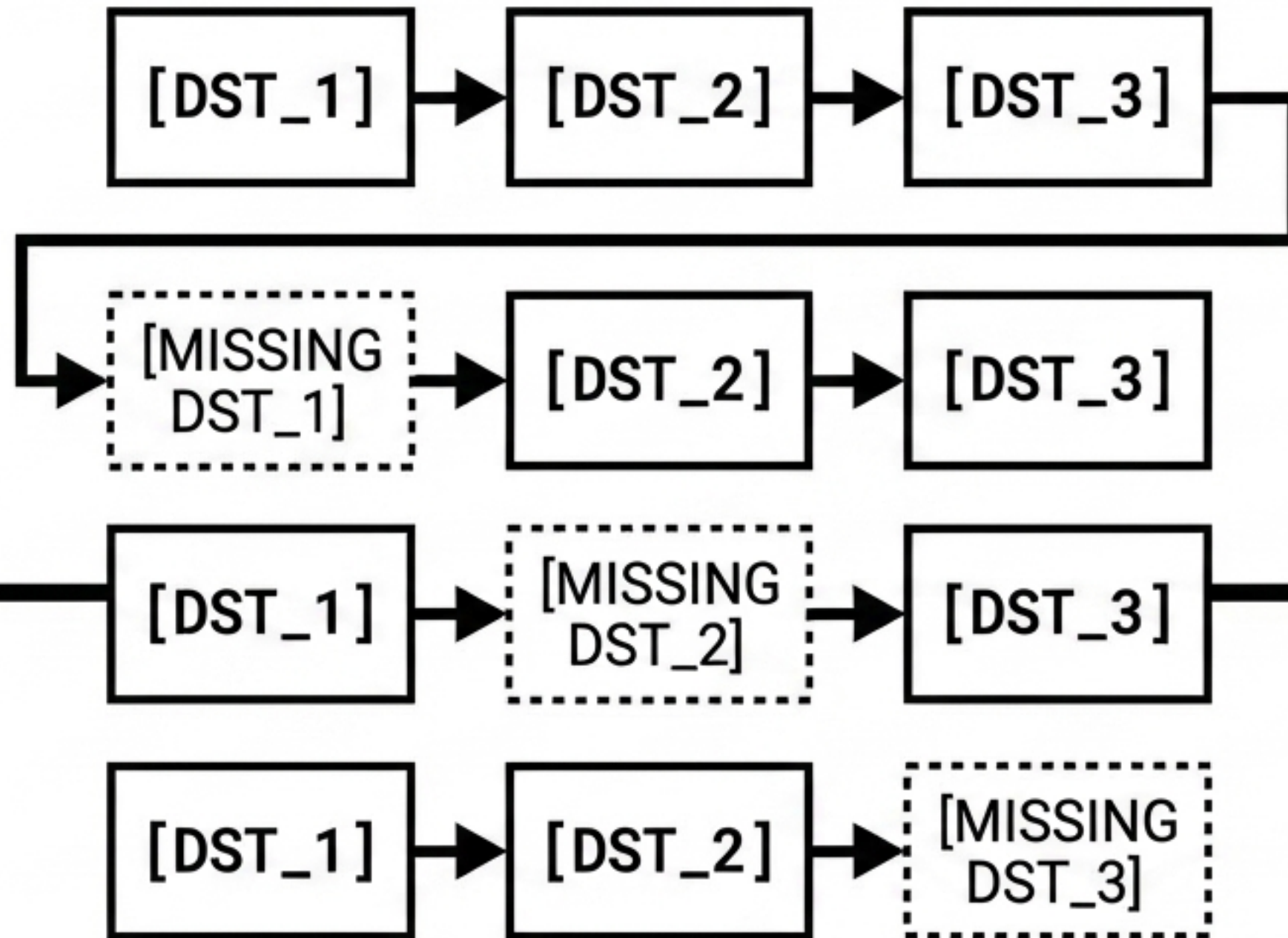
Test 2: Sufficiency



$< \theta_{\text{sufficiency}}$

PASSED: A is Sufficient

Minimality Ablation and Statistical Rigor



Systematic Ablation: Hunting
for Minimal Sufficient Set

STATISTICAL VALIDATION CONSTRAINTS:

Bootstrapped C.I.:
1000 iterations

Type I error bound:
 $\alpha = 0.05$

Confidence constraint:
> 95%

Causal Insight Graph Synthesizer (CIGS)

S2

Terminal Effect Node
(Churn Risk)

S1

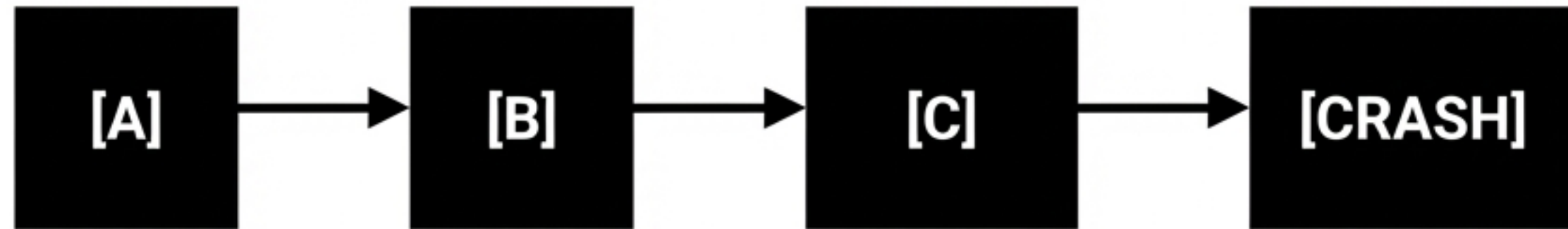
Propagation Node
(Latency Shift)

S0

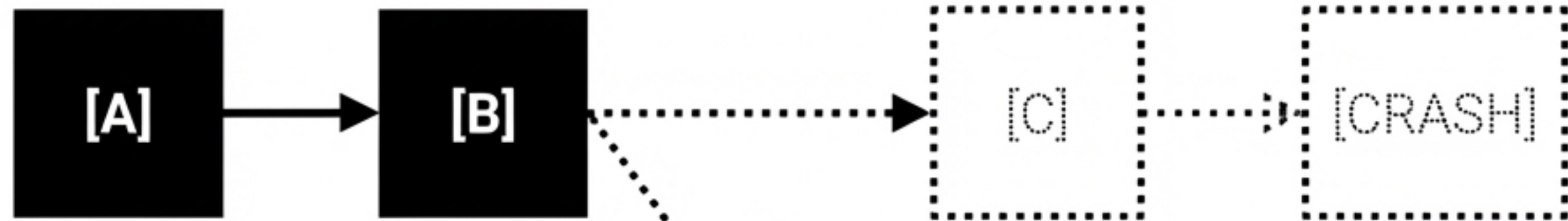
Root Cause DST
(Firmware_v3.2)

Predictive Precursor Engine (PPE)

Historical Memory:
Precursor Sequence
Library (PSL)



Live Stream
Observation

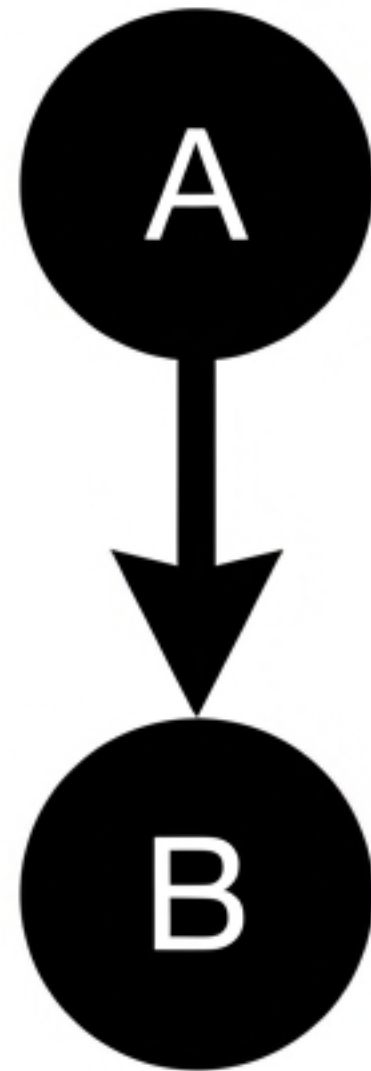


$P(\text{incident} \mid \text{prefix})$

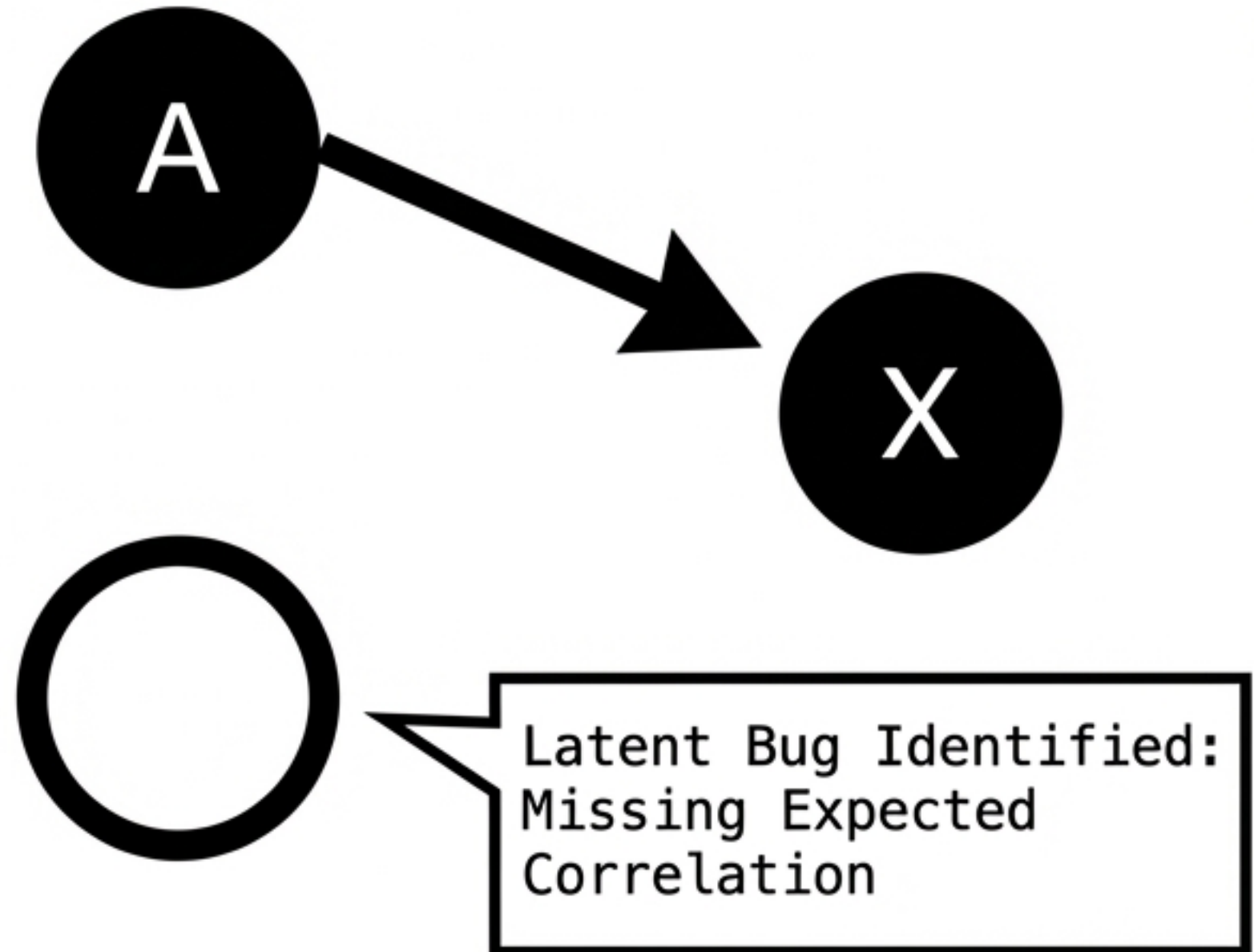
The AMACE Anti-Correlation Paradigm

Detecting Missing Expected Correlations

Expected Path



Observed Path



The Computational Leap

Prior Art / Naive Search

$$O(N^2 \times K^D)$$

- Combinatorial explosion
- High false positive rate
- Coincidental correlations

The AMACE Pipeline

$$O(N \times \log(N) \times k \times \log(D))$$

- Entropic pruning
- Bounded Type I errors
- Counterfactually validated

Observable Outputs & Defensibility

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}
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